

Catching Hallucinations in Bengali LLM Answers

Settle what you can prove, then ask the models

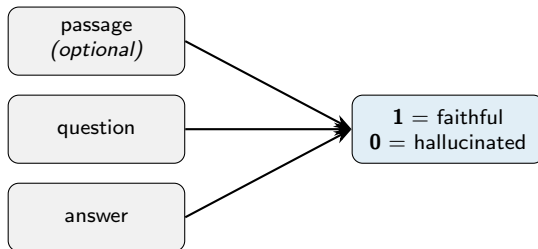
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The task

We are given three things. We must return one number.



299 labelled rows to learn from. 2,516 test rows to predict. Only 299 — so every design choice has to be cheap to validate.

A wrong answer is a few edits away from a right one

Question	Faithful	Hallucinated
National poet of Bangladesh	Nazrul	Tagore
Where oral saline was developed	Bangladesh (icddr,b)	the West
Which Nobel Yunus won	Peace	Economics
Father of the nation	Bangabandhu	Gandhi

These are the **C1 cultural-distance** rows — where the Bangladeshi answer differs from the global default, and the competition's tie-breaker.

A general multilingual model gets them wrong **systematically**, not randomly.

Two very different jobs

Whether a passage is supplied changes the task completely.

Track	The question we ask	Labelled	Test	Our CV
Grounded	Is it supported by the passage?	130	1,361	0.946
Closed-book	Is it true in the world?	169	1,155	0.743

Our encoder scores AUC 0.972 on grounded rows but 0.571 on closed-book — barely better than a coin toss. So it only votes on grounded rows.

Every threshold, judge weight and combiner is fitted separately per track.

We classified the test set before building anything

Eight recurring question types, detected from prompt/response structure only — never from labels. Ordered by **how much we trust the resolver**.

#	Question type	How we detect it
1	Reference-answerable QA	prompt found in our verified corpus
2	Known hallucination pair	(prompt, answer) matches a verified pair
3	Cultural-default trap	matches a hand-authored Bangladeshi pattern
4	Idiom field swap	idiom found; literal vs figurative field
5	Mathematical reasoning	operator / math-keyword regex
6	Grounded numeric claim	answer states a number absent from context
7	Duplicate prompt	same normalised pair inside the test set
8	Residual factual QA	<i>none of the above matched</i>

Types 1–4 are settled by evidence, 5–7 by a rule, and type 8 is what the models are for.

Types 1–4: settled by verified evidence

We assembled 104,874 verified Bengali question–answer references from public sources, plus an 11,051-idiom dictionary.

#	What we check	Verdict	Precision
1	Answer contains a verified gold answer	faithful	0.988
2	Answer matches a verified wrong answer	hallucinated	1.000
3	Answer is a known-wrong global default	hallucinated	validated
4	Answer gives the <i>other</i> meaning field	hallucinated	1.000

Sources: squad_bn, BanglaRQA, BanglaQuAD, NCTB textbook QA, public exam sets, BenHalluEval.

All public, all cited, **none derived from the test set**. Where no exact match exists, the gold answer is instead shown to the judge for meaning comparison.

Type 4 in detail: the idiom swap-trap

Bengali idiom questions ask for *either* the literal (**shabdik**) *or* the figurative (**bhabartho**) meaning.

Asked field	Response gives	Verdict
figurative meaning	the figurative meaning	faithful
figurative meaning	the <i>literal</i> meaning	hallucinated

The distractor is a **correct** meaning of the right idiom — just the wrong field. A generic semantic-similarity judge scores it as a near-perfect match and passes it.

We load both meanings per idiom and check against the **asked** field. 1.000 precision on 116 test rows.

Types 5–7: settled by a rule

These need no world knowledge at all. A short program decides them.

#	The rule	Precision
5	Re-solve the expression with SymPy and compare	1.000
6	Answer's number is absent from the passage $\Rightarrow$ wrong	1.000
7	Identical rows must receive identical labels	consistency

Every layer is precision-gated

A layer switches on only if it clears **0.90 precision** on the labelled set with enough support, and disables itself if its data fails to load.

That leaves **type 8** — the residual. It has no shortcut, and it is what the three judges are for.

Context injection from external corpora

For closed-book rows we have no passage. So we **retrieve one** and give it to the judge.

Corpus	Content	Why
bn-Wikipedia	general knowledge	broad factual coverage
XL-Sum	BBC Bangla news	recent and named entities
bdlaws	Bangladeshi statutes	legal and civic facts

306,000 indexed chunks, BM25 retrieval, then the row is re-judged **with the retrieved passage injected** into the prompt.

Gated deliberately hard. Ungated, loose lexical retrieval on trivia is near-random — it caused one of our two score regressions. It now fires on 121 rows only.

This is the component we expect to **grow** on the held-out fold.

Three judges, chosen to disagree usefully

Judge	What it brings	Read as
TigerLLM-9B	Bengali-native fluency and culture	logit of hyan vs na
Qwen3.5-9B (bn)	multilingual breadth	logit of yes vs no
Qwen3.5-9B (en)	English-instruction frame	logit of yes vs no
BanglaBERT	fine-tuned encoder, grounded only	classifier probability

Why the English frame? A multilingual model often judges more reliably when the instruction and verdict space are English while the content stays Bengali — it [recovers Bangladeshi facts it knows better in English space](#). Worth +0.011.

BanglaBERT is fine-tuned on [synthetic squad_bn contrastive pairs](#) — each answerable squad_bn question yields a faithful gold answer and a minimally-perturbed hallucinated one — plus real benchmark pairs; squad_bn prompts in the labelled set are held out to keep its gate honest.

All 4-bit NF4. Tiger on GPU 0, Qwen on GPU 1, running concurrently.

Reading a verdict without generating text

We do not ask the model to write an answer. We read the **first-token logit** of the yes/no verbalizer from a single forward pass.

	Generate the verdict	Read the logit
Tokens per row	many	one forward pass
Output	text to parse	a calibrated probability
Relative cost	baseline	$\approx 10\times$ cheaper

This gives us a *number* rather than a word, which is what makes the per-track blending and threshold tuning possible at all.

Prompts are two-shot and register-matched, and warn the model that wrong answers are usually a slightly altered name, year or number.

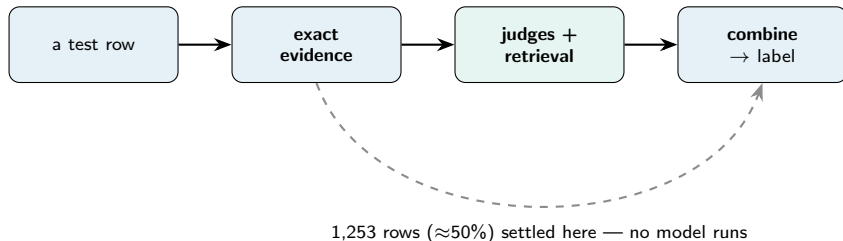
Combining them, separately per track

Track	Combiner chosen	Threshold	CV macro-F1
Grounded	logistic-regression stack	0.35	0.946
Closed	three-way blend (Qwen-heavy)	0.75	0.743

- Three candidate combiners are cross-validated per track. A challenger is adopted only if it beats the simple blend by more than 0.005.
- Out-of-fold predictions keep the stack honest on only 295 rows.
- Maths rows are excluded from the closed fit — SymPy overrides them anyway.
- Thresholds come from nested CV. **Nothing is hand-tuned to the leaderboard.**

Putting it together

Cheap, reliable checks run first. Models only decide what is left over.



Layers are applied in **ascending order of trust**, so better-validated evidence wins ties. If evidence and a model disagree, **the evidence wins**.

Every tier fire is written to `match_audit.csv` — every deterministic decision is traceable.

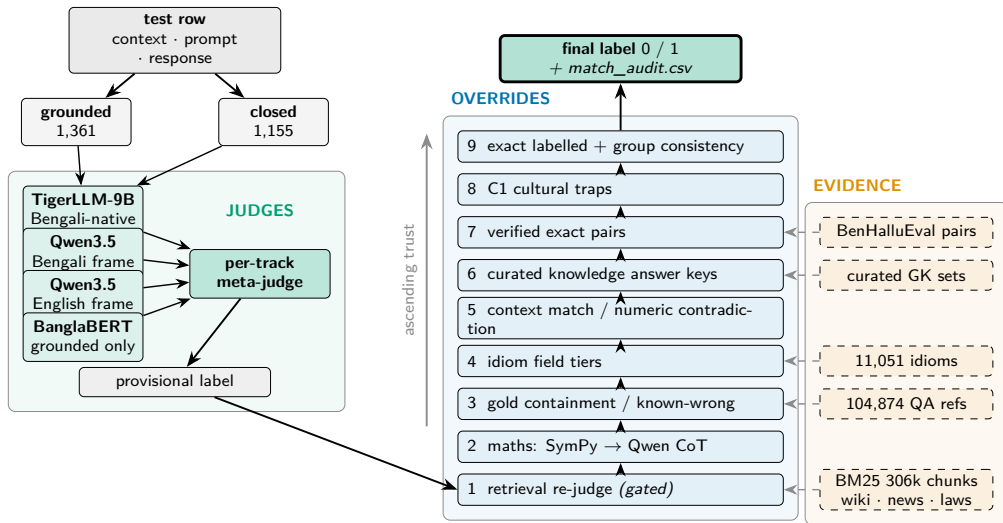
The exact decision order

Later means higher trust. The test-time pipeline mirrors the labelled simulation one-for-one.

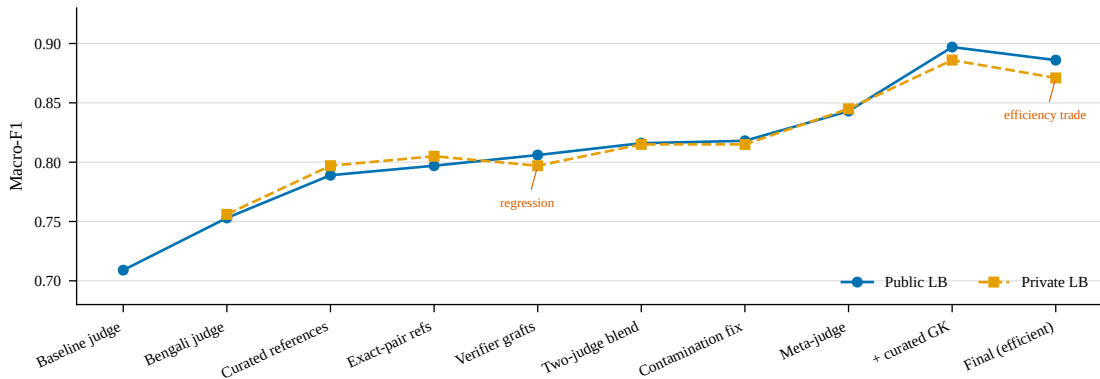
Step	Layer
1	Per-track meta-judge probability $\rightarrow$ threshold
2	Retrieval re-judge on confident closed rows
3	Qwen-only base for maths rows (Tiger removed)
4	SymPy override, then Qwen chain-of-thought where SymPy abstains
5	Gold containment $\rightarrow$ 1 known-wrong $\rightarrow$ 0
6	Idiom field tiers
7	Context containment $\rightarrow$ 1 numeric contradiction $\rightarrow$ 0
8	Curated knowledge tiers, then verified exact pairs
9	C1 cultural traps
10	Group consistency across duplicate rows

Final test distribution: 54.2% faithful / 45.8% hallucinated — balanced, with **no verdict collapse**.

The Whole Pipeline



Ten submissions, two of them backwards



The biggest single model win was replacing a general multilingual judge with a **Bengali-native** one. Public split is ≈ 750 rows, so deltas under 0.01 are noise.

The two dips taught us the most

Dip 1 — a checker that had already seen the answers

Our QA verifier's gate reported 0.971 precision. But its encoder was fine-tuned on squad_bn's train split, and 86% of our validation rows come from squad_bn — it was measuring **memorisation**. On harder unseen rows it certified hallucinations as faithful.

Standing rule: before trusting any gate, ask **what its model was trained on**.

Dip 2 — a learned decider that won CV and lost the leaderboard

A gradient-boosted decider scored 0.852 in simulation and **0.776** on the board.

Standing rule: gate on **flip precision**, never on CV deltas alone. In-sample simulations flatter.

What we tried and rejected

Measured, then dropped. We do not revisit these without new evidence.

Rejected approach	Measurement
Learned decider replacing the threshold	won CV, lost 2 LB points
Distractor pairs as encoder training data	closed AUC fell to 0.558
Qwen3-14B <i>replacing</i> the 8B judge	0.880 vs 0.897; OOMs on T4
Ungated BM25 retrieval on trivia	caused the 0.784 regression
Chain-of-thought for factual judging	hurts 12 of 21 benchmark combos
14B maths solver	≈580 s/row, ate the budget
Fully-synthetic hallucination set	invented entities, never matches

A stronger judge that *replaces* a weaker one can lose net signal — it only helps as an **added** member, and we had no T4 headroom for that.

Engineering traps we hit at least once

Trap	Fix
Bengali mixes Unicode composition forms	NFC-normalise everything, keys included
Projecting a 262k vocab OOMs a T4	logits_to_keep=1, eager attention
Qwen3 emits <think> first	enable_thinking=False
Test lacks a column the harness injects	guard it; it crashed a run
Kaggle mounts inputs at nested paths	resolve by content, not hardcoded globs
A dataset can silently fail to attach	check the banner in the first two minutes

The silent-attach failure is the dangerous one: ≈ 900 rows quietly fall back to the judge and cost about 0.03 if nobody reads the log.

Did it work?

Configuration	Grounded	Closed
Tiger alone	0.828	0.709
Best two-way blend	0.838	0.737
Selected per-track meta-judge	0.946	0.743
Full pipeline, all layers	0.870 macro-F1	0.855 F1(label=0)

Leaderboard	Macro-F1
Public	0.886
Private	0.871

Offline simulation 0.870 vs private 0.871. The system **generalises rather than memorises**.

Why these models, and not others

The organisers published a benchmark. We let it choose for us. **Lower is better.**

Model	Factual QA	Reasoning
GPT-4.1-mini (closed API)	15.56	26.40
DeepSeek-R1-14B	38.60	18.10
Qwen2.5-32B	46.85	26.80
Gemma-2-27B	47.45	39.05
LLaMA-3.1-8B	47.71	53.30
Mistral-nemo-12B	53.84	49.30
TigerLLM-9B	28.28	53.45

Second best of seven on facts, beating 27–32B models $\Rightarrow$ our main judge.

Worst of seven on reasoning — it misses 85.5% of reasoning hallucinations $\Rightarrow$ we banned it from maths.

Scores from BenHalluEval (Adib et al., IUT) — the organisers' own benchmark.

Making it finish inside the rerun budget

Change	Before	After
Main judging block	≈ 241 min	≈ 73 min
Encoder training per run	≈ 18 min	0 (frozen checkpoint)
Generation batching	$1\times$	$\approx 7\times$
Rows already settled	still judged	skipped entirely
Verdict method	generated text	single-pass logit

Two T4s, 4-bit NF4, under 50 GB of disk, well inside the 9-hour limit. Chunk-level parquet checkpoints mean an interrupted run resumes exactly.

We gave up ≈ 1.5 private-leaderboard points for a package that provably completes — the rerun is 50% of the grade.

What survives the held-out fold, and what does not

The Phase-2 fold draws from broader sources and contains **zero** rows from the datasets our exact-match tiers key on.

Component	Phase-1 value	Phase-2
squad_bn gold-answer matching	≈41% of rows	dead by design
Verified exact pairs	≈900 rows	dead by design
Three-judge blend	decides residuals	≈90% of the score
Retrieval from external corpora	gated, advisory	grows
Encoder	grounded only	yes, grounded
Idiom field tiers	116 rows	keyed on idioms, not data
Numeric + paragraph checks	130+ rows	register-generic

We knew this in advance and chose the architecture accordingly — which is why we shipped the **efficient** notebook, not our highest-scoring one.

What we have not solved

- **Closed-book is the ceiling.**

≈ 0.74 CV. World-knowledge factuality without evidence is genuinely hard, and it is where the durable headroom is.

- **299 labelled rows is very little.**

Several tiers auto-disabled for want of support — by design, but it means some ideas remain untested rather than disproven.

- **Reference coverage is finite.**

51% of test prompts matched our corpus. The rest rests entirely on the neural judges.

The honest reading

Roughly half our Phase-1 margin comes from evidence that will not exist on the held-out fold. The transferable core is **judges, retrieval, dictionaries and numeric verification** — and that is where the next work goes.

Three things to take away

#	Lesson	Evidence
1	Settle what you can verify before you ask a model	worth +0.06
2	Grounded and closed-book are two tasks, not one	AUC 0.972 vs 0.571
3	Let a benchmark pick your model for each task	28.28 vs 53.45

The benchmark told the field *which* Bengali models detect hallucinations well on *which* tasks.

We turned that measurement into a working, efficient, precision-gated **detector** — and it transfers.

0.886 public · **0.871** private · offline simulation **0.870**

Thank you

Questions?

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